

Assigment 7

Hilal Selen Bütün - Batuhan Denizli - Ramazan Karagöz

```
install.packages("remotes")
remotes::install_github("htastan/TRmaps")
install.packages("TRmaps")
install.packages("sf")
install.packages("dplyr")
install.packages("ggplot2")
library(TRmaps)
library(sf)
library(dplyr)
library(ggplot2)

library(TRmaps)
library(sf)
library(dplyr)
library(ggplot2)

data("tr_nuts3")

df_divorce <- read.csv("divorce.csv", header = TRUE, sep = ";", dec = ",")

df_divorce_final <- df_divorce |>
  rename(name_tr = 1, Kaba_Bosanma_Hizi = last_col()) |>
  mutate(name_tr = gsub("\\(.*\\)", "", name_tr),
         name_tr = trimws(name_tr)) |>
  filter(name_tr != "Türkiye", !is.na(Kaba_Bosanma_Hizi)) |>
  mutate(Kaba_Bosanma_Hizi = as.numeric(Kaba_Bosanma_Hizi)) |>
  filter(!is.na(Kaba_Bosanma_Hizi)) |>
  select(name_tr, Kaba_Bosanma_Hizi)

tr_divorce_map <- left_join(tr_nuts3, df_divorce_final, by = "name_tr")

ggplot(tr_divorce_map) +

  geom_sf(aes(fill = Kaba_Bosanma_Hizi), color = "gray60", size = 0.1) +

  geom_sf_text(aes(label = name_tr),
              size = 1.8,
              color = "black",
              fontface = "bold",
              check_overlap = FALSE) +

  scale_fill_distiller(
    palette = "Blues",
    direction = 1,
```

```

name = "Number of divorces per
1000 population (2024)"
) +

labs(
  title = "Crude Divorce Rate by Province",
  subtitle = "Number of divorces per thousand population",
  caption = "Source: TUIK"
) +

theme_void() +
theme(
  legend.position = "right",

  legend.title = element_text(size = 8, face = "bold"),
  legend.text = element_text(size = 7),

  plot.title = element_text(hjust = 0.5, face = "bold", size = 16, margin = margin(b = 5)),
  plot.subtitle = element_text(hjust = 0.5, size = 11, color = "gray30"),
  plot.caption = element_text(size = 7, color = "gray50", margin = margin(t = 5)),

  plot.margin = unit(c(0.2, 0.1, 0, 0.1), "cm")
) +

guides(fill = guide_colorbar(
  barheight = unit(3.5, "cm"),
  barwidth = unit(0.3, "cm"),
  frame.colour = "black",
  ticks.colour = "black"
))

```

Crude Divorce Rate by Province

Number of divorces per thousand population



Source: TUIK